

SSGMCE	SHRI SANT GAJANAN MAHARAJ COLLEGE OF ENGG., SHEGAON - 444 203.		LABORATORY MANUAL	
	PRACTICAL EXPERIMENT INSTRUCTION SHEET			
	EXPERIMENT TITLE:	TO CREATE LIST AND PERFORM VARIOUS OPERATIONS ON IT.		
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Aim: To create list and perform various operations on it.

Facilities: Computer with RStudio installed.

Theory:

In R, the variables are not declared as some data type. The variables are assigned with R-Objects and the data type of the R-object becomes the data type of the variable. There are many types of R-objects. The frequently used ones are:

1. Vectors
2. Lists
3. Matrices
4. Arrays
5. Factors
6. Data Frames

Vector: A vector is simply a list of items that are of the same type.

List: A list is an R-object which can contain many different types of elements inside it like vectors, functions and even another list inside it.

Matrix: A matrix is a two-dimensional rectangular data set. It can be created using a vector input to the matrix function.

Array: While matrices are confined to two dimensions, arrays can be of any number of dimensions. The array function takes a dim attribute which creates the required number of dimension.

Factor: Factors are the r-objects which are created using a vector. It stores the vector along with the distinct values of the elements in the vector as labels. The labels are always character irrespective of whether it is numeric or character or Boolean etc. in the input

vector. They are useful in statistical modeling.

Data Frame: Data frames are tabular data objects. Unlike a matrix in data frame each column can contain different modes of data. The first column can be numeric while the second column can be character and third column can be logical. It is a list of vectors of equal length.

Operations on list:

Create List: To create a list, use the **list()** function:

Ex:- mylist <- list("apple", "mango", "grape")
print(mylist)

Access List: We can access the list items by referring to its index number, inside brackets. The first item has index 1, the second item has index 2, and so on:

Ex:- print(mylist[2])

Change Item Value: To change the value of a specific item, refer to the index number:

Ex:- mylist[3] <- "banana"
print(mylist)

List Length: To find out how many items a list has, use the **length()** function:

Ex:- length(mylist)

Check If Item Exist: To find out if a specified item is present in a list, use the **%in%** operator:

Ex:- "watermelon" %in% mylist

Add List Items: To add an item to the end of the list, use the **append()** function:

Ex:- append(mylist, "orange")
print(mylist)

To add an item to the right of a specified index, add **"after=index number"** in the **append()** function:

Ex:- append(mylist, "orange", after = 2)
print(mylist)

Remove List Items: We can also remove list items. The following example creates a new, updated list without an "apple" item:

Ex:- newlist <- mylist[-1]


```
print(newlist)
```

Range of Indexes: We can specify a range of indexes by specifying where to start and where to end the range, by using the : operator:

Ex:- `print(mylist[2:3])`

Loop through a List: We can loop through the list items by using a **for** loop:

```
Ex:- for (x in mylist) {  
  print(x)  
}
```

Join Two Lists: There are several ways to join, or concatenate, two or more lists in R. The most common way is to use the `c()` function, which combines two elements together:

```
Ex:- list1 <- list("a", "b", "c")  
list2 <- list(1,2,3)  
list3 <- c(list1,list2)  
print(list3)
```

Conclusion: We have successfully performed various operations on list.